LAB : 4

Title: switch statement and flow control in cpp

► Objective

The purpose of this lab is to:

- Understand and use the switch statement in C++.
- Apply the break statement to control program flow.
- Use the continue statement inside loops.
- Differentiate between break and continue.
- Develop simple menu-driven programs.

► Theory

1. Switch Statement

• The switch statement is used to select one block of code from multiple options based on the value of a variable.

• It is an alternative to multiple if-else statements when checking a single variable.

• Each case in a switch corresponds to a constant integral expression.

• The default case handles unmatched values.

Syntax:

```
switch (expression) {
    case value1:
        // code
        break;
```

```
    case value2:
        // code
        break;
    default:
        // code
}
```

2.Break Statement

- Ends the execution of the current case.
- Prevents fall-through into subsequent cases.
- Without, execution continues until the next break or the end of the switch block.

3.Continue Statement

- Used inside loops.
- Skips the current iteration and moves to the next iteration of the loop.

► Programs and Implementation

❖ Program 1: Day of the Week

```
#include <iostream>        // tells the compiler to include input output library
using namespace std;    // tells the compiler to use standard namespace

int main() {                                        // main fuction starting point
    int day;                                        // variable declaration
   cout << "Enter day number (1-7): ";      // output promt for user
```

```
cin >> day;                                   // takes input from user

switch(day) {                          // switch statement  which switches day
    case 1: cout << "Monday"; break;                    // case 1 monday
    case 2: cout << "Tuesday"; break;                   // case 2 tuesday
    case 3: cout << "Wednesday"; break;                 // case 3 wednesday
    case 4: cout << "Thursday"; break;                  // case  4  thursday
    case 5: cout << "Friday"; break;                    // case 5 friday
    case 6: cout << "Saturday"; break;                  // case 6 saturday
    case 7: cout << "Sunday"; break;                    // case 7 sunday
    default: cout << "Invalid day!";                  // default invalid day

return 0;                         // tells the computer that program has ended
  }
```

Sample output:

Enter day number (1-7): 3 Wednesday

❖ Program 2: Grade Evaluation

```
#include <iostream>       // tells the compiler to include input output library
using namespace std;     // tells the compiler to use standard namespace

int main () {                                       // main fuction starting point
     char grade;                                    // variable declaration
  cout << "Enter grade (A, B, C, D, F): ";         // prompt for user
       cin >> grade;                                // takes input from user
```

```
switch(grade) {                                   // switch statement which switches grade
    case 'A': cout << "Excellent"; break;             // case A excellent
    case 'B': cout << "Very Good"; break;            // case B very good
    case 'C': cout << "Good"; break;                 // case C good
    case 'D': cout << "Pass"; break;                 //   case D pass
    case 'F': cout << "Fail"; break;                 // case f fail
    default: cout << "Invalid Grade!";            // default invalid grade
  }
      return 0;                         // tells the computer that program has ended
  }
```

Sample output:

```
Enter grade (A, B, C, D, F): B very good
```

❖ **Program 3: Conversion Menu**

```
#include <iostream>      // tells the compiler to include input output library
using namespace std;    // tells the compiler to use standard namespace

int main() {                    // main fuction starting point
    int choice;                // variable declaration
    double value;              // variable declaration

   cout << "Menu:\n";                                      /*   promt for
   cout << "1 -> Kilometers to Meters\n";                   user
   cout << "2 -> Meters to Centimeters\n";              imput
```

```
cout << "3 -> Kilograms to Grams\n";            what conversion
cout << "4 -> Celsius to Fahrenheit\n";         you   want       */

cout << "Enter your choice: ";          // prompt for user to enter choice
cin >> choice;                          // takes input from user

switch(choice) {                        // switches choice

  case 1:                                  /*   case 1
    cout << "Enter kilometers: ";             message to display
    cin >> value;                               takes value
    cout << "Meters = " << value * 1000;      conversion   and
    break;                                      output to display  */

  case 2:                                       /* case 2 for conversion
    cout << "Enter meters: ";                    cout for user
    cin >> value;                                takes value from user
    cout << "Centimeters = " << value * 100;     converts and outputs
    break;                                       value and the break */
  case 3:                                  /* case 3
    cout << "Enter kilograms: ";             prom pt for user
    cin >> value;                            takes value from user
    cout << "Grams = " << value * 1000;      converts and displays
    break;                                   value to display and break */

  case 4:                                       /* case 4
```

```
        cout << "Enter Celsius: ";                          prompt for user
        cin >> value;                                         takes value from user
        cout << "Fahrenheit = " << (value * 9/5) + 32;      converts and display
        break;                                                 value to display */

      default:                                /* default
        cout << "Invalid choice!";            outputs invalid choice message  */
          }
       return 0;                      // tells the computer that program has ended
       }
```

Output sample:

```
Menu:
1 -> Kilometers to Meters
2 -> Meters to Centimeters
3 -> Kilograms to Grams
4 -> Celsius to Fahrenheit
Enter your choice: 4
Enter Celsius: 25
Fahrenheit = 77
```

► Results

- The switch statement successfully selects among multiple options.
- The break statement prevents fall-through and ensures correct program flow.
- The default case handles invalid inputs.
- Menu-driven programs can be efficiently implemented using switch.

► Conclusion

In this lab, we learned:

- How to implement switch-case statements in C++.
- The importance of break in controlling execution.
- The role of default in handling unmatched cases.
- Practical applications of switch in building calculators, grade evaluators, and conversion menus.